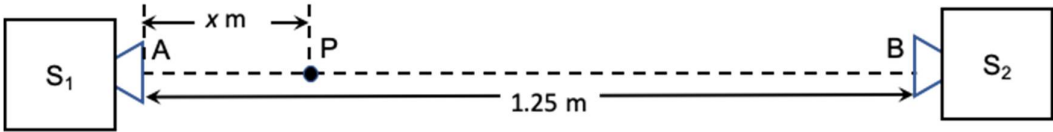


			[1]
5	(a)	State two differences between a progressive wave and a stationary wave.	
		difference 1	

		 difference 2	
			[2]
		Any two of the following (1 mark each): • The amplitude in a stationary wave varies from zero to a maximum ($2A$) at the antinode, whereas the amplitude is the same for all points in a progressive wave. • Between any 2 adjacent displacement nodes, all points in a stationary wave are always at the same phase of oscillation; points in adjacent nodal loops are always oscillating in antiphase. However, for a progressive wave, all points within one wavelength are in different phases of oscillation. • Waveform of a stationary wave does not advance, whereas the waveform of a progressive wave advances (with the velocity of the wave). • No energy is transferred along a stationary wave, whereas energy is transferred from point to point in the direction of travel of the progressive wave.	
	(b)	Two coherent radiowave sources S_1 and S_2 that are in phase face each other, separated by a distance of 1.25 m, as shown in Fig. 5.1. Point P is located along line AB, x metres away from A.  Fig. 5.1	
	(i)	Show that the positions of minimum intensity along the line AB can be described by the following equation: $2L = (2n+1)\lambda$ where L is the path difference between the waves from S_1 and S_2, n is any integer and λ is the wavelength of the radiowaves.	
			[1]
		Since the 2 sources are in phase, for locations where minimum intensity is located, the path difference between the 2 waves must be an odd number integer of half wavelength. $L = (2n+1) \frac{\lambda}{2}$ $2L = (2n+1)\lambda$	

		(ii)	Determine the number of points along AB where minima would be expected when radiowave signals of wavelength 0.429 m are produced.	
			number of points =	[3]
			<u>Method 1</u> $1.25 / 0.429 = 2.9 \text{ wavelengths}$ $= 5.8 \text{ half-wavelengths}$ → 6 nodal points <u>Method 2</u> $2L = (2n+1)\lambda$ $2(1.25 - 2x) = (2n+1)\lambda$ $2.5 - 4x = (2n+1)\lambda$ $2.5 - 4x = (2n+1)(0.429)$ $x = \frac{(2n+1)(0.429) - 2.5}{-4}$ Since $0 \leq x \leq 1.25$ $0 \leq \frac{(2n+1)(0.429) - 2.5}{-4} \leq 1.25$ $-5 \leq (2n+1)(0.429) - 2.5 \leq 0$ $-2.5 \leq (2n+1)(0.429) \leq 2.5$ $-3.41 \leq n \leq 2.41$ Therefore, $n = -3, -2, -1, 0, 1$ and 2. There are 6 positions. <u>Method 3</u> $2L = (2n+1)\lambda$ $2(1.25 - 2x) = (2n+1)\lambda$ $2.5 - 4x = (2n+1)\lambda$ $2.5 - 4x = (2n+1)(0.429)$ $x = \frac{(2n+1)(0.429) - 2.5}{-4}$ Substitute $n = 0, 1, 2, -1, -2, -3, \dots$ until x cannot be more than 1.25 m, or less than 0 m. There will only be 6 possibilities. Mark scheme for Methods 2 & 3: 1 mark: correct derivation of expression for x 1 mark: correct calculations of x for the substitutions 1 mark: final answer (n has 6 possibilities)	